

Bence Hervay

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Education

Computer Science BA — University of Cambridge, Trinity College

Oct 2021 – Jun 2024

Second Best Computer Science Dissertation (92.5%) — Out of all 120+ students

Full Scholarship — Awarded to only 5 students based on academic merit

Achieved strong 2:1 overall

Advanced Mathematics Class — Fazekas Mihály Grammar School

Sep 2015 – Jul 2021

1st in Mathematics and Physics National Olympiads (2018 and 2020); Consistently top 10 (2017 – 2021)

Robotics World Champions — FIRST Global Challenge (2017); 150+ countries; White House reception

25+ national top-3 placements across Mathematics, Computer Science, Physics, and Robotics (2016 – 2021)

Achieved A*A*A* equivalent final grades with a Medallion of Excellence for outstanding performance

Artificial Intelligence Technology Training — KÜRT Academy

Jan 2019 – Jun 2019

Completed as the youngest trainee ever (at the age of 16)

Work Experience

Quantitative Research Engineer — AGITProp

Sep 2024 – Present

Automatic detection and statistical refinement of financially interpretable factors for covariance matrix prediction

Primary ownership of numerous key modules in the pipeline (mainly mathematical components)

Key role in project planning and coordination, contributing to firm-level prioritisation decisions

Constant expansion and maintenance of the infrastructure (e.g. quantitative modules, data pipeline, testing)

Software Developer Intern — DRW

Jun 2023 – Sep 2023

Created a fully automatic monitoring and resource allocation tool interacting with 200+ employees daily

Utilised NLP techniques to filter and classify bug reports mentioned in messages for efficiency

Quantitative Research Intern — WorldQuant

Jun 2022 – Aug 2022

Generalised index rebalance alphas by automatic inference of rebalance schedules and methodologies

Developed a framework to mass-produce low-correlation alphas by training decision forests

Senior Group Leader — Alphacademy

Aug 2021 – Jun 2023

Member of the founding team; interviewed, onboarded, and supervised 20+ teachers

Taught programming to 100+ students from novice to Olympiad level

Projects & Research

Rupert's Problem — C++ & TypeScript

Jun 2024 – Present

Settled the central open conjecture of a 350-year-old mathematical topic — whether every convex polyhedron fits through itself

Heavily computer-assisted proof involving 10,000+ lines of C++ utilising interval arithmetic to bound floating-point errors

Largest independent research project with 1,000+ hours invested; comprehensive preprint in preparation

Reinforcement Learning for 2048 (Cambridge Dissertation) — C++

Sep 2023 – May 2024

Developed the world's highest-scoring algorithm and the fastest simulation engine for the 2048 Puzzle Game

Reaches the 2048-tile in 1.2 milliseconds with a 99.8% success rate and 75% of the time even the 32768-tile

Trained a custom 200M-parameter RAMnet via self-play reinforcement learning on 150M games

Combined expectimax with locally brute-forced critical phases to maximise the probability of survival

Selected Additional Projects

Built without relying on high-level libraries to ensure deep understanding of the fundamentals

GM-level chess engine — Awarded a research scholarship by the National Research Excellence Program

3D Path Tracer — Won the Cambridge rendering competition organised held across the full cohort

Project Euler — Top 0.04% of all users by solving each of the first 222 problems (~250 in total)

C++ automatic differentiation framework — Used for a custom video assistant referee

Stacked MWPC for muon tomography — Invited to the Large Hadron Collider (CMS detector)

Handwritten digit classifier DNN with a GUI; Rubik's cube solver with a GUI; Time-travel based platformer game

Solvers for Multiplayer Pac-Man, Sudoku, Nonogram, and numerous other strategic games

Skills

Python & C++ with 200,000+ LOC experience, quantitative and statistical methods, ML, advanced DSA, low-level optimisation

Cross-disciplinary STEM expertise; data-driven solutions to theoretical and practical problems from first principles

Hobbies & Sports

Rubik's Cube speedsolving — 14-second average; 20x national top 20

Ultimate Frisbee — UK National Championship 6th (2023)

Great Court Run — 3rd (2021); 2nd (2022); 1st and Beat the Clock (2023)

Acrobatic Basketball World Cup 4th (2018) — FaceTeam Youngsters

Activities

Regular instructor at the European Summer Program on Rationality

Events Officer of Cambridge University Hungarian Society

Captain of Trinity College Ultimate Frisbee Society

Always happy to discuss interesting puzzles and brain teasers